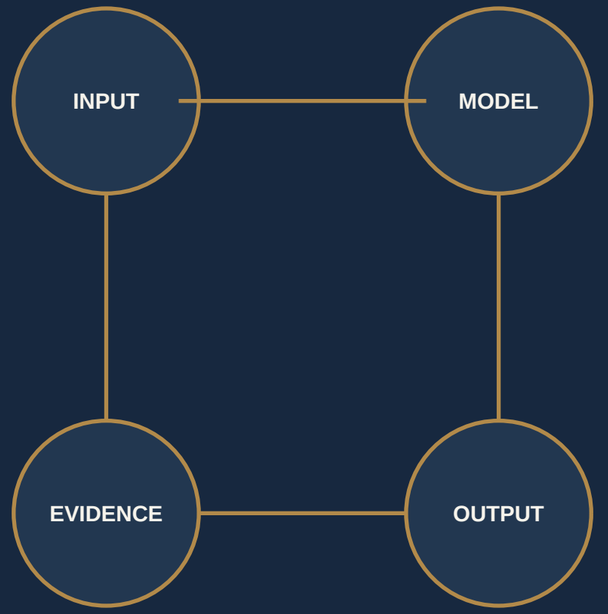
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robotics Fundamentals

AI.102 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define the fundamental principles of robotics.
- Apply key methods including robot anatomy.
- Build or analyze a practical system involving coordinate systems.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

The fundamental principles of robotics

02

METHODS

Robot anatomy

03

SYSTEMS

Coordinate systems

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · The fundamental principles of

The fundamental principles of robotics

02 · Robot anatomy

Robot anatomy

03 · Coordinate systems

Coordinate systems

04 · Actuators

Actuators

05 · Sensors

Sensors

06 · Motors

Motors

07 · Gears

Gears

08 · Mechanical structures

Mechanical structures

Core ideas, made explicit

AI.102 • Distinguish definitions, assumptions and evidence

- The fundamental principles of robotics
- Robot anatomy
- Coordinate systems
- Actuators
- Sensors

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.102 • Distinguish definitions, assumptions and evidence

- Motors
- Gears
- Mechanical structures
- End-effectors
- Robot controllers

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
The fundamental principles of robotics

INPUTS
Robot anatomy

METHOD
Coordinate systems

OUTPUT
Actuators

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving the fundamental principles of robotics.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using the fundamental principles of robotics.
- Implement or simulate a baseline using robot anatomy.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.